Method 3: angle θ_S equals the angle between the ray tangents at the observer:

$$\sin\frac{\theta_S}{2}=\frac{\triangle(r\cos\varphi)}{\triangle r}=\cos\varphi-r\sin\varphi\frac{\triangle\varphi}{\triangle r}\ . \qquad (11)$$

From the ray equation (1), calculate $\triangle\varphi/\triangle r$:

$$\sin\frac{\theta_S}{2}=\cos\varphi-r\sin\varphi\frac{GM/c^2}{r^2a\sin\varphi}=\cos\varphi-\frac{GM}{c^2ra}\ ;$$

Insert observer coordinates $r=d$, $\phi=-\pi/2$ and a from (4), noting small θ_S:

$$\theta_S\approx\frac{2GM}{c^2d}\sqrt{\frac{2c^2d}{GM}}=\sqrt{\frac{8GM}{c^2d}},$$

This agrees with (6), again giving (8).

Method 4: use (2) to rewrite (1) as

$$-r_m=r\cos\varphi-\frac{GM}{c^2r_mr}\left[(r\cos\varphi)^2+2(r\sin\varphi)^2\right],$$

Thus $$x=-r_m+\frac{GM}{c^2r_mr}(x^2+2y^2)\ . \qquad (12)$$

For $y\to-\infty$, the asymptote of (12) is

$$x=-r_m-\frac{2GM}{c^2r_m}y\ .$$

This straight line has x intercept $-r_m$ and slope

$$\frac{1}{-2GM/c^2r_m}\approx\frac{1}{-\tan(\theta_S/2)}\approx-\frac{1}{\theta_S/2}\ .$$

Thus $\theta_S\approx 4GM/c^2r_m$. Substitute (5) for r_m to obtain (6), and hence (8).

7.

1. (I) Helium has two electrons; E^+ is the energy to remove one to infinity, satisfying $He\quad E^+\to He^++e^-$. For E^+, find the ion energy after one electron is ionized, E^*: one electron orbiting the helium nucleus. Here are two methods.

Method 1: in the mechanical equation

$$\frac{2ke^2}{r^2}=\frac{mv^2}{r}$$

r is the radius and v the speed. Bohr's ground-state quantization ($\hbar$) gives

$$r_0=\hbar^2/2ke^2m\ . \qquad (1)$$

Thus the helium-ion energy is

$$E^*=\frac{p_0^2}{2m}-\frac{2ke^2}{r_0}=-\frac{2k^2e^4m}{\hbar^2}\ , \qquad (2)$$